# Robert Burdsall

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## Summary

Electrical Engineering student at NC State with hands-on experience taking production hardware from concept to commissioning: control panel design and wiring, sensor and machine vision integration, PLC machine control, and embedded robotics software. Led an automated inspection machine running 120 parts/min end to end and deployed first-time automated defect detection on 3 production machines. Seeking an avionics, test, or manufacturing engineering internship.

## Education

**North Carolina State University** — Raleigh, NC
*Bachelor of Science in Electrical Engineering, GPA: 3.5* — *Expected Dec. 2028*

- **Relevant Coursework:** Introduction to Signals, Circuits and Systems; Fundamentals of Logic Design; Computer Systems Programming; Introduction to Computer Systems

**North Carolina School of Science and Mathematics** — Morganton, NC
*High School Diploma, Computer Science Focus, GPA: 4.8* — *Aug. 2023 – May 2025*

## Experience

**IDEAL Fastener Corporation** — Oxford, NC
*Electrical Engineering Intern* — *May 2026 – Aug. 2026*

- Owned an automated inspection machine end to end — concept, electrical design, build, and commissioning — running 120 parts/min with Keyence machine vision, vacuum ejection, solenoid valve sequencing, a DOPSoft HMI, and PLC quality tracking with good/bad rate alarms.
- Designed and fabricated 5+ electrical control panels, performing full wiring and harnessing across relays, power supplies, DIN rail, disconnects, proximity and laser sensors, and solenoid valves.
- Produced TinyCAD schematics for every panel so plant technicians could replicate, service, and troubleshoot the hardware from documentation alone.
- Established automated defect detection for the first time on 3 existing production machines, reaching 100% inspection coverage by configuring Keyence IV3/IV4 vision systems and defining defect recognition criteria.
- Programmed Omron PLCs in CX-Programmer, implementing ladder logic control sequences, stepper motion pulse output, and buffered part tracking through the machine.

**FIRST Robotics Competition, Team 9150 Labyrinth Robotics** — Morganton, NC
*Software Developer (2024–25); Lead Programming Mentor (2025–Present)* — *Aug. 2024 – Present*

- Designed and implemented the full robot control software in Java using WPILib, integrating vision and dual scoring systems on hardware built and debugged under a compressed build-to-competition schedule.
- Achieved the 3rd-highest scoring performance across 2025 North Carolina events, including the UNC Asheville District Event.
- Lead and mentor 25+ students, teaching Java, WPILib, wiring, and iterative test and debug of a competition robot.

**North Carolina School of Science and Mathematics** — Morganton, NC
*Computer Science Teaching Assistant* — *Aug. 2024 – May 2025*

- Taught 150+ students programming fundamentals in Java and Python.
- Built a cross-platform resource tracking application in Flutter and Dart in collaboration with faculty.

## Projects

**Aurum** | *Dart, Flutter, Python, Express.js, TensorFlow* — Sep. 2024 – Dec. 2024

- Built a cross-platform mobile application for real-time coin appraisal, running on-device inference with TensorFlow Lite.
- Developed the Express.js backend with JSON APIs and JWT authentication for live value retrieval and portfolio tracking.

## Organizations

**Boy Scouts of America, Troop 222** — Huntersville, NC
*Eagle Scout; Senior Patrol Leader and Assistant Scoutmaster* — *2015 – Present*

- Mentor 30+ Scouts as Assistant Scoutmaster and lead outdoor programs focused on leadership, teamwork, and problem-solving.

## Skills

**Hardware and Test**: Control panel design and fabrication, wiring and harnessing, sensor integration (proximity, laser), machine vision (Keyence IV3/IV4, IV Smart Navigator), solenoid and pneumatic actuation, PLC programming (Omron CX-Programmer), HMI development (DOPSoft), schematic capture (TinyCAD)
**Design and Fabrication**: SolidWorks (Essentials for Part Design certificate), 3D printing (Prusa Slicer, Cura)
**Languages**: Java, Python, C/C++, Dart, SQL, JavaScript, HTML/CSS
**Developer Tools**: Git, Docker, VS Code, IntelliJ, TensorFlow, Google Cloud Platform, Android Studio, Xcode